Eco-Health-Viz User Manual

A guide to exploring synthetic county-level data

Open Eco-Health-Viz online · Developed by Mohammad Shaito, Ph.D.

1. Choose the records to explore

Open **Configuration and filtering** in the sidebar. Choose a **State-labeled synthetic scenario**. Each scenario contains 50 fictional counties and 36 monthly records per county, from January 2021 through December 2023.

Pick both endpoints of the **Date range**. The app includes month-end records that fall within those dates. Leave **County subset** empty to include all counties, or select the fictional counties you want to compare.

The **?** beside each control explains its purpose. On a phone, tap the sidebar icon at the top left to open or close the controls.

2. Choose a view

Environmental and disease comparison. Pick simulated PM2.5 (µg/m³), temperature offset (°C), or household income (nominal units) for the horizontal axis. Pick simulated COVID-19 or influenza cases per 100,000 people per month for the vertical axis. Each point represents a county and month. Hover over a point for details. County colors appear for selections of up to ten counties.

Monthly trends. The upper panel shows disease rates. The lower panel shows average PM2.5 across the selected counties. The panels share a month axis and use separate vertical scales.

Data inspection and export. View up to 100 filtered rows in the table. Use the download controls to save the full filtered dataset and its provenance.

3. Interpret patterns

The simulator applies an annual seasonal cycle to environmental variables and expected disease rates. A visible association can arise from this shared timing. Use the charts to explore the behavior of the synthetic scenarios.

Rates, downloads, and troubleshooting

4. Compare monthly rate summaries

Pooled population rate. For each month, divide total simulated cases by the total population of the selected counties, then multiply by 100,000. More populous counties carry more weight.

Mean county rate. Calculate each county rate per 100,000 and average those rates. Each county carries equal weight.

PM2.5. The lower panel averages the selected counties with equal weight in each month.

5. Save a dataset and its settings

In **Data inspection and export**, select **Download filtered CSV** to save all matching records. The CSV includes simulated case counts and rates alongside the expected values used by the generator.

Select **Download provenance JSON** with the same filters. Keep it with the CSV so you can identify the scenario, dates, counties, software versions, and checksums.

6. Resolve common issues

No observations. Select both date endpoints and include at least one month-end date. For example, select January 1 through December 31, 2022 for twelve months of data.

Unexpected county selection. Check the selected scenario. Its county picker lists fictional counties for that scenario. Clear the selection to restore all 50 counties.

Missing download controls. Switch **Analysis mode** to **Data inspection and export**.

7. Run locally

Open a terminal in the folder containing `app.py` and `requirements.txt`. Run these commands:

```
python -m pip install -r requirements.txt
streamlit run app.py
```

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